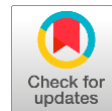


Application of machine learning algorithms for NPL prediction based on financial data. The case of Albania



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Abstract Machine learning has become increasingly important in predicting nonperforming loans (NPLs), offering greater accuracy than traditional risk assessment methods. As lending activity continues to expand in Albania, accurately forecasting nonperforming loans is therefore essential to minimize potential risks to financial stability. Using data obtained from the Bank of Albania, this study aims to forecast the NPL ratio for the period 2017–2023 using seven bank-specific financial indicators. Two supervised learning algorithms: Random Forest (RF) and Support Vector Machine (SVM) are employed for this purpose and model performance was evaluated using root mean squared error (RMSE) and mean absolute error (MAE). Furthermore, this paper identifies key factors via feature importance analysis and evaluates how excluding less relevant variables impacts predictive accuracy. The results show that before feature selection, RF model outperformed the SVM, achieving the lowest RMSE and MAE values of 0.015 and 0.012, respectively. The most influential variables were identified as *Loan Loss Reserves/Total Loans* and *Bank Loans/Total Assets*, whereas several of the indicators exhibited negligible effects. Model performance for both RF and SVM improved slightly after removing less relevant variables, as indicated by lower RMSE and MAE values. While the SVM model's accuracy increased after dimensionality reduction, RF model still outperformed it overall based on the performance metrics. Specifically, the simplified RF achieved superior results, recording an RMSE of 0.011 and an MAE of 0.009, while also attaining the highest adjusted R^2 among all models. The findings confirm the importance of variable selection in enhancing model performance and demonstrate RF as the most effective tool for NPL prediction in Albania. These results provide practical implications for banks and regulators, supporting the development of early warning systems and contributing to financial stability.

Keywords: nonperforming loans, random forest, support vector machine, feature selection

1. Introduction

Lending is undoubtedly the main activity of retail banking. While most loans are repaid in full and on time, some default when borrowers fail to meet their repayment obligations. These are commonly referred to as nonperforming loans (NPLs). In recent years, European regulators have paid close attention to NPLs, as many banks still face difficulties disposing of those that accumulated on their balance sheets during the financial crisis. (Bellotti et al., 2021) Financial stability and innovation are fundamental drivers of a nation's economic health and long-term development. The banking system faces significant risk when NPLs increase, potentially undermining financial stability. High levels of NPLs imply broader macroeconomic challenges, such as constrained credit supply, slower economic growth, and declining investor confidence. (Arnone et al., 2024)

Therefore, continuous monitoring and in-depth analysis of NPLs are crucial for maintaining them at manageable levels. In Albania, the number of loan applications reviewed by banks in 2024 increased by 12.1% compared with the previous year. As a result, preventing these loans from turning into defaults becomes increasingly significant, as they have a direct impact on national economic performance. (Nerjaku & Sinaj, 2024)

Global economic activity has continued to expand. As observed both internationally and in Albania during the second half of 2024, the banking sector developed sustainably, characterized by expanding deposits and loans, improved profitability, and controlled risk levels. Positive economic developments—including increased consumption, investment, employment, and wages, as well as an eased monetary policy—supported higher credit demand from businesses and households. The acceleration of lending was accompanied by a marginal increase in the stock of nonperforming loans (NPLs); however, the NPL ratio remained at historically low levels. Specifically, the NPL ratio has continued to decline, reaching its lowest value in the last 15 years at 4.2%, driven by faster growth in outstanding loans, repayments of obligations by clients, and write-offs from bank portfolios. Overall, economic stability and prudent banking policies contributed to reducing credit risk and sustaining steady credit growth, thereby maintaining the resilience of the banking sector despite external challenges. (Bank of Albania, 2024)

Given the growing volume and complexity of financial data, the integration of machine learning (ML) and artificial intelligence (AI) in loan default prediction has become increasingly essential, demonstrating superior precision and adaptability compared with traditional methods. ML techniques are designed to enhance predictive accuracy, particularly in contexts where the underlying data-generating process is complex or unknown. (Coulombe & Surprenant, 2019) They are widely adopted in macroeconomic and financial analyses, as evidenced by their widespread application across numerous empirical findings. (Abdullah et al., 2023) apply machine learning techniques to predict nonperforming loans in banks across 15 emerging countries. Among all 14 linear and nonlinear models, the random forest model outperforms the other models in forecasting bank nonperforming loans, achieving 76.10% accuracy. Additionally, the empirical findings of (Zhu et al., 2019) suggest that random forest (RF) provides superior predictive accuracy compared with logistic regression, decision trees and several other ML algorithms in default prediction. RF is one of the most extensively tested ML algorithms across a wide range of predictive modeling tasks. In many comparative studies, such as Annisa & Rusdah (2022), Kalayci (2017), Sinap (2024) and Chaturvedi et al. (2025), the random forest has outperformed traditional models and several advanced algorithms. The RF model performed better than the support vector machine (SVM) model in the papers of (Gatera et al., 2023; Ji, 2024). Moreover, the findings of (Nerjaku & Sinaj, 2024) showed that SVM achieved the highest predictive performance among random forests and artificial neural networks (ANNs) in predicting NPL based on key macroeconomic indicators such as GDP, the consumer price index, and the unemployment rate in Albania. Furthermore, the SVM outperforms the RF, even in the paper of (Syahputra & Wibowo, 2023). In the study of (Azeta et al., 2023), an interactive machine learning framework was developed using a support vector machine to predict nonperforming loans. The SVM model achieved 96% accuracy, outperforming other algorithms and confirming its superiority in loan classification tasks. The SVM has demonstrated strong predictive performance even in the results of (Setiawan et al., 2022).

The systematic review of (Noriega et al., 2025) highlights the need for financial institutions to increase the use of artificial intelligence (AI) and machine learning (ML) to assess credit risk by analyzing large volumes of information. Furthermore, the findings of (Chang et al., 2024) indicate that accurate credit risk analysis supports better-informed lending decisions, and the use of machine learning and deep learning algorithms has notably enhanced predictive performance in this field.

The level of NPLs is a critical indicator for both banks and policymakers, as it reflects the overall health of the financial sector. The primary objective of this research is to highlight the role of machine learning techniques in forecasting the NPL ratio. Specifically, we aim to compare the effectiveness of two methods—random forest and support vector machine—in predicting the NPL ratio. Furthermore, this paper identifies the key factors influencing the models through a feature importance analysis and evaluates the extent to which feature-based variable selection enhances their predictive accuracy. By evaluating the performance of each model, this study seeks to determine which approach yields more accurate predictions, thereby providing valuable insights for financial institutions and regulatory bodies.

2. Materials and methods

This paper aims to predict nonperforming loans in Albania via seven independent bank-specific variables, as presented in Table 1. In addition to macroeconomic factors, the results of (Sinay et al., 2022) show that bank-specific factors have a direct effect on NPL. Furthermore, the empirical findings of (Alnabulsi, 2022) and (Gafsi, 2025) indicate that the level of nonperforming loans is more sensitive to bank-specific factors than macroeconomic factors.

Table 1 Description of the dataset.

Variable	Description
X1	Loan loss reserves/Total Loans
X2	Net income/Total Assets (ROA)
X3	Net income/Total Equity (ROE)
X4	Bank Loans/Total Assets
X5	Total Deposits/Total Assets
X6	Equity/Total Assets
X7	Log (Total Assets)
Dependent variable	Non-Performing Loans (NPL)

The dataset consists of Bank of Albania's quarterly observations for the years 2017-2023. (These data are available at <https://www.bankofalbania.org>) For the modeling process, two machine learning models, Random Forest and Support Vector Machines were employed to evaluate and compare predictive accuracy. The root mean square error (RMSE) and mean absolute error (MAE) were used as performance metrics. The RMSE is computed as the square root of the mean squared difference between the predicted ($\hat{y}_i$) and actual values (y_i). The RMSE calculates the average predictive error via the following formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

The MAE calculates the mean of the absolute differences between the predicted ($\hat{y}_i$) and actual values (y_i) as represented below:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

RF is a robust, nonlinear ensemble learning algorithm designed to handle large datasets with multiple features. It operates by constructing an ensemble of decision trees, each trained on a bootstrap sample of the data with random feature selection. These trees generate independent predictions, and the final output is obtained through majority voting in classification tasks or averaging in regression tasks. This mechanism not only enhances the predictive accuracy but also mitigates the risk of overfitting associated with individual decision trees. Moreover, the random forest method demonstrates strong performance in the presence of missing values, outliers, and noisy data, making it a practical and widely adopted approach for real-world applications. (Breiman, 2001) (Patel et al., 2020)

One of the most popular supervised machine learning algorithms based on statistical learning theory is the support vector machine (SVM). It uses a kernel function, usually nonlinear, to map input data into a high-dimensional space and looks for the best hyperplane between the two classes. This separation is achieved by maximizing the margin between the closest data points, known as support vectors, and the decision boundary (Pisner & Schnyer, 2020). Support Vector Regression (SVR) extends the SVM's to continuous outputs by introducing an ϵ -insensitive tube around the regression function. Specifically, this is formulated as a convex optimization task by minimizing an ϵ -insensitive loss function and identifying the flattest possible tube that captures most training points. The resulting unique solution expresses the regression hyperplane in terms of support vectors, those samples that fall outside the ϵ -tube's boundaries (Awad & Khanna, 2015).

3. Results and Discussion

First, NPL prediction was conducted via seven input variables and two supervised predictive models: a random forest model and a support vector machine. Model performance was evaluated with the RMSE and MAE, with the results summarized in Table 2 and represented graphically in Figure 1.

Table 2 Comparison of the RF and SVM algorithms in terms of the RMSE and MAE.

Model	RMSE	MAE
RF	0.01512921	0.012092551
SVM	0.02552162	0.016193935

As observed, the random forest outperforms the support vector machine in terms of both the RMSE and the MAE, achieving approximately an RMSE of 0.015 and an MAE of 0.012.

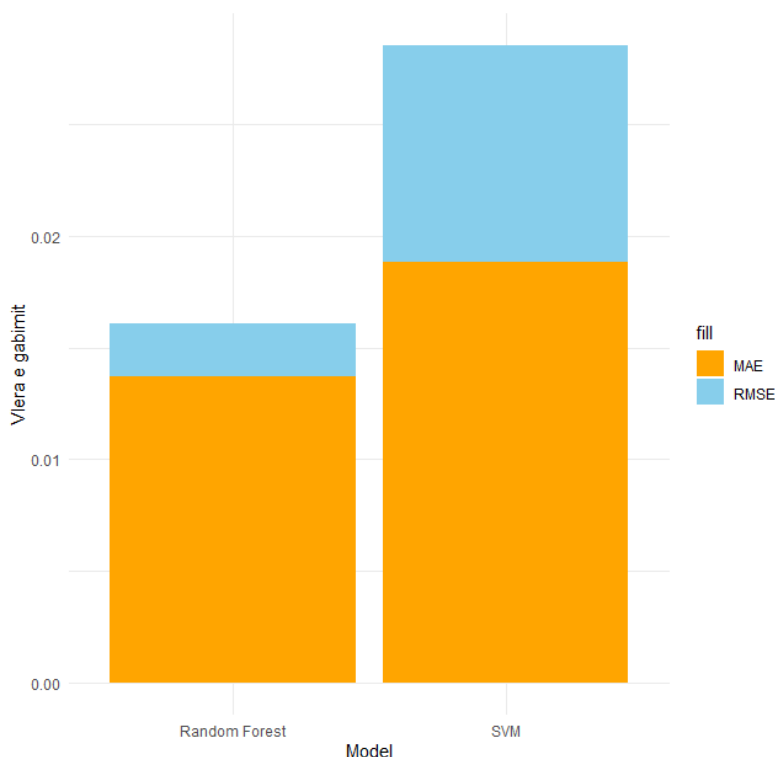


Figure 1 Comparison of Model Performance.

(Dewi & Chen, 2019) demonstrated that selecting the most relevant features is essential for improving model performance. RF has proven to be an effective model for feature selection, as it highlights the most important variables and helps identify those with limited relevance, as noted by (Chen et al., 2020). Figure 2 illustrates the results of the variable importance analysis, clearly indicating that loan loss reserves/total loans (X1) and bank loans/total assets (X4) hold the highest predictive values. On the other hand, ROA (X2), ROE (X3) and equity/total assets (X6) display values close to zero on both axes, suggesting that their impact on model performance is negligible.

The empirical analyses of (Alessi et al., 2021) and (Chukwuogor et al., 2021) further confirm the strong relationship between NPL and LLR. The results of (Karadima & Louri, 2019) also confirm that Bank Loans/Total Assets is statistically significant to the NPL ratio. Moreover, the findings of (Irawan & Syarif, 2019) report an insignificant relationship between Bank Loans/Total Assets and NPL. Furthermore, the insignificant relationship between ROA and ROE with NPLs is supported by the findings of (Ranabhat & Subedi, 2022) but contradicted by the study of (Jiang & Zheng, 2024), who reported strong relationships between ROA, ROE and NPLs. Similarly, the insignificant effect of Equity/Total Assets on NPLs is consistent with the findings of (Rachman et al., 2018).

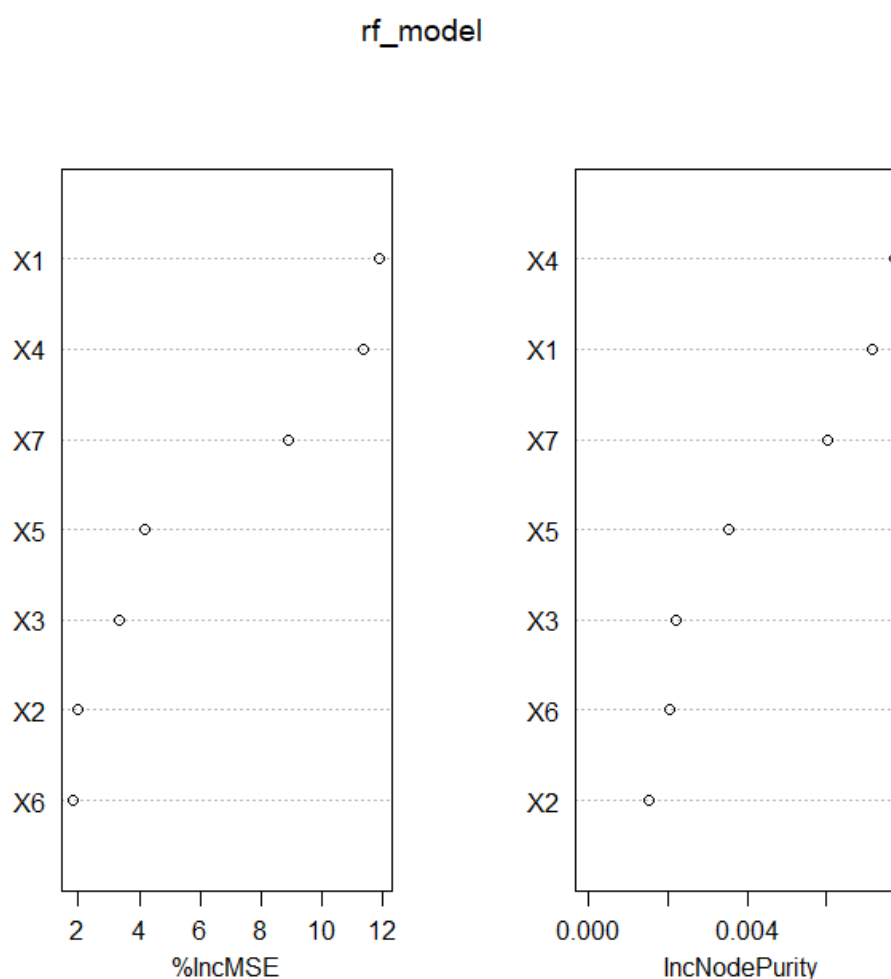


Figure 2 Feature importance analysis. RF model.

Consequently, only the four most important variables were retained in the second random forest model, whereas ROA, ROE, and equity/total assets were excluded because of their low significance. Both the RMSE and MAE values slightly decreased after the less important variables (X2, X3, and X6) were eliminated, suggesting improved model performance, as shown in Table 3 and Figure 3. This illustrates a well-known feature of random forest models: greater numbers of variables do not always lead to better outcomes (Hastie et al., 2009). Conversely, concentrating on the most significant features tends to simplify the model structure and enhance the predictive accuracy.

Table 3 Comparison of RFs before and after feature selection on the basis of the RMSE and MAE.

Model	RMSE	MAE
Random Forest (full)	0.01512921	0.012092551
Random Forest (simplified)	0.01101819	0.009381762

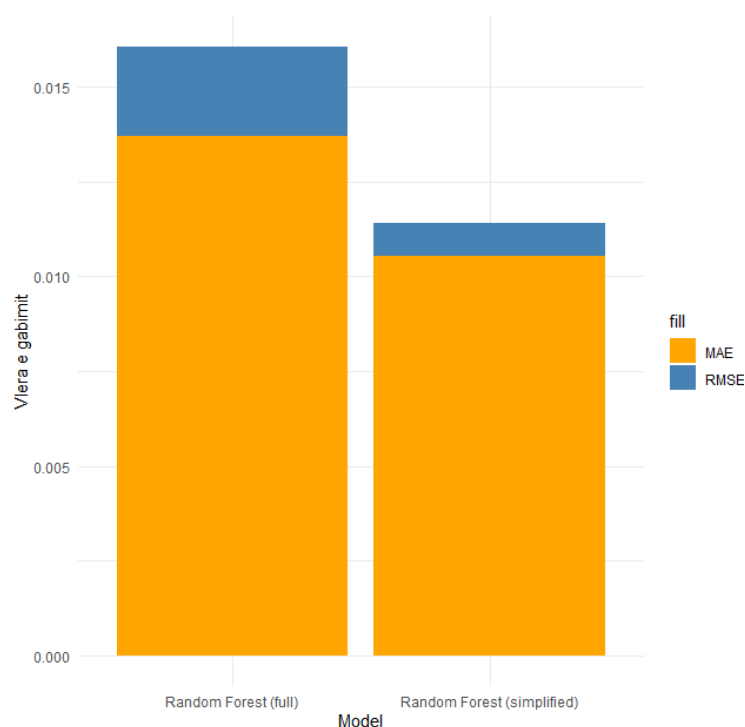


Figure 3 Comparison of random forest model performance before and after feature selection.

Following the feature importance analysis with SVM, again, the variables Loan loss reserves/Total Loans (X1) and Bank Loans/Total Assets (X4) emerged as the most significant contributors to predictive performance, as illustrated in Figure 4. In contrast, variables X5 and X6, corresponding to Total Deposits/Total Assets and Equity/Total Assets, were excluded from the SVM model because of their negligible importance and minimal contribution to predictive performance. The revised SVM model, which uses only the five most relevant features (X1, X2, X3, X4, X7), was subsequently re-evaluated via the RMSE and MAE.

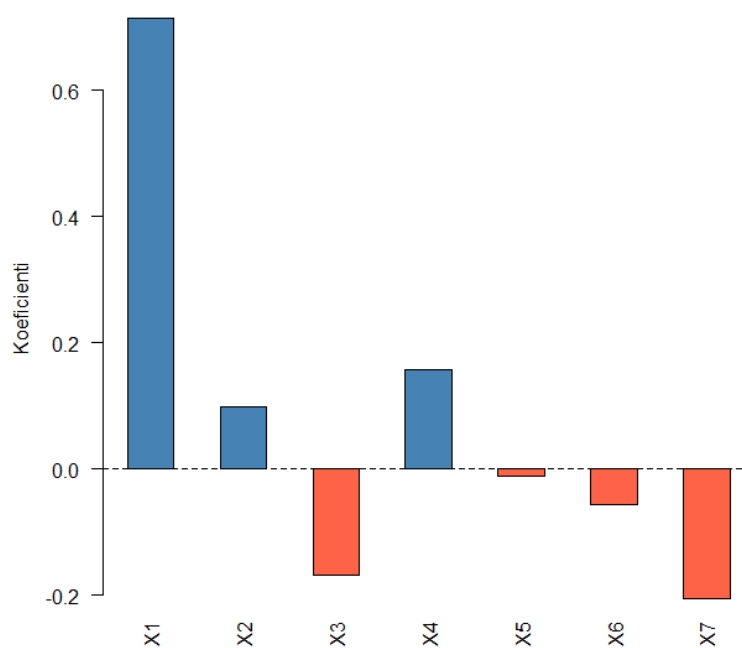


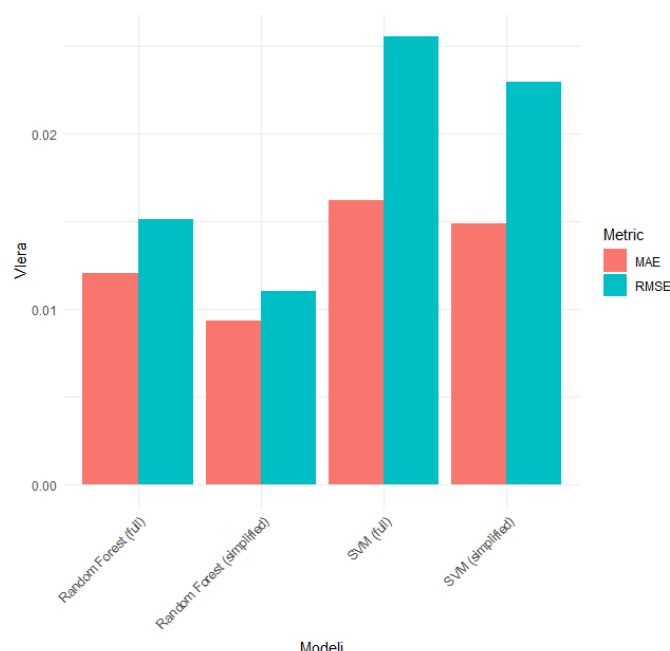
Figure 4 Feature importance analysis of the SVM model.

The results showed a slight improvement compared with the initial SVM model, confirming that even for support vector machines, reducing the input dimensionality can enhance model accuracy and reduce noise. However, the overall performance was still lower than that of the random forest model. RF superiority over SVM is further supported by the results of (Adugna et al., 2022). The results of all four models are shown in the table below (Table 4) and represented graphically in Figure 5.

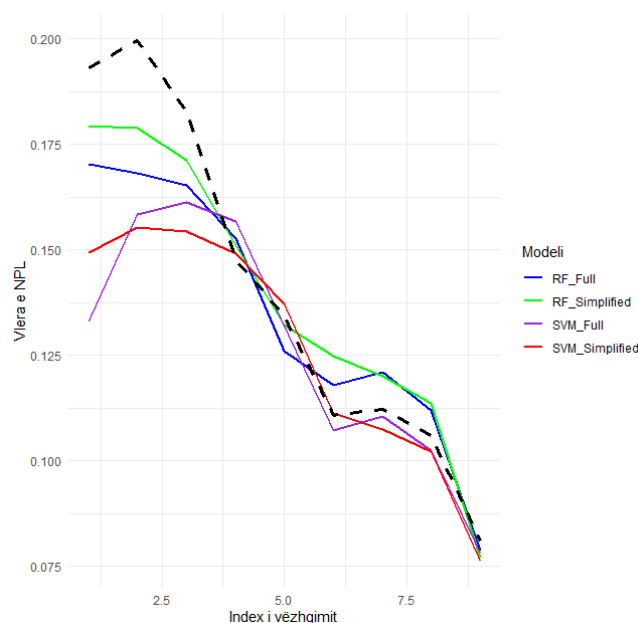
Table 4 Comparison of the RF and SVM methods, both before and after feature elimination, on the basis of the RMSE and MAE.

Model	RMSE	MAE
Random Forest (Full)	0.01512921	0.012092551
Random Forest (Simplified)	0.01101819	0.009381762
SVM (Full)	0.02552162	0.016193935
SVM (Simplified)	0.02295473	0.014920628

As shown in Table 4, feature selection has a significant effect on improving outcomes. This result is consistent with the findings of (Cheng, 2025). Compared with all the other models, the random forest (Simplified) model achieves the lowest RMSE ≈ 0.011 and MAE ≈ 0.009 , clearly outperforming both the random forest (Full) and the SVM in both the pre- and postfeature selection scenarios.

**Figure 5** Comparison of the RF and SVM methods, both before and after feature elimination, on the basis of the RMSE and MAE.

In conclusion, the simplified random forest model was found to be the most effective approach for predicting nonperforming loans in this context. As shown in Figure 6, which compares the predicted and actual values, the simplified RF model produces predictions that are significantly closer to the actual line, indicating a superior model fit.

**Figure 6** Comparison of each model prediction with actual data.

This result is further confirmed by the minimal loss observed in Figure 7 and the highest adjusted R^2 value achieved, as shown in Figure 8.

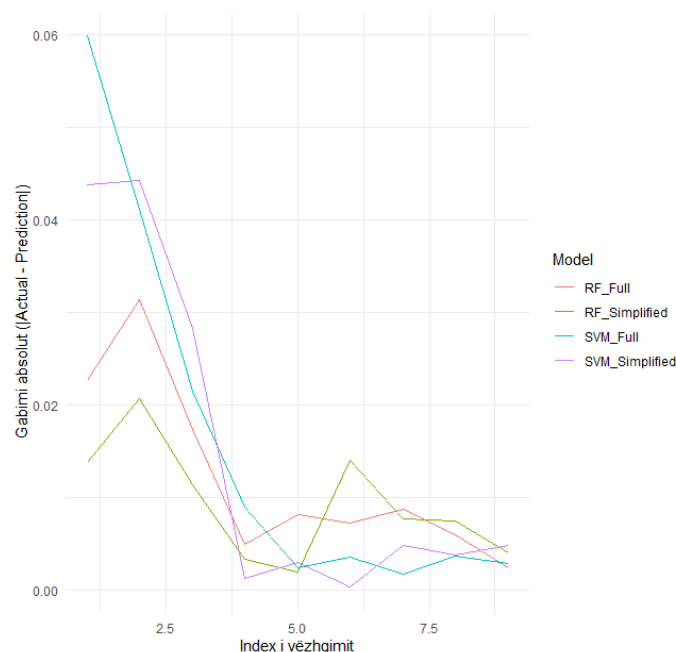


Figure 7 Comparison of the absolute loss for each model on the test set.

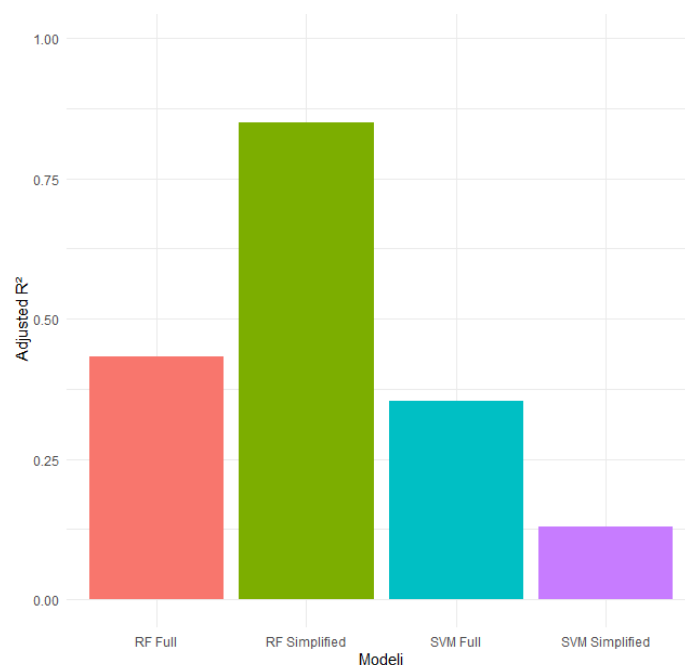


Figure 8 Comparison of adjusted R^2 values for all the models.

4. Conclusions

This study demonstrates that the random forest outperforms the support vector machine in predicting nonperforming loans in Albania on the basis of bank-specific financial indicators. The analysis identified *Loan Loss Reserves/Total Loans* and *Bank Loans/Total Assets* as the most critical determinants of NPLs, whereas the other variables contributed minimally. Feature selection not only reduces model complexity but also improves predictive accuracy, making the simplified RF model the optimal approach in this context.

From a practical perspective, the results suggest that financial institutions and the Bank of Albania can integrate RF-based models into their risk monitoring systems to develop more reliable early warning mechanisms. This would enable earlier identification of potential problem loans and support preventive measures to safeguard financial stability.

Nonetheless, some limitations must be acknowledged. The study relies solely on bank-specific indicators over a relatively short period (2017–2023). Future research could incorporate macroeconomic factors, extend the dataset, and test advanced machine learning methods such as gradient boosting or neural networks. Additionally, applying cross-validation techniques would strengthen the robustness and generalizability of the results.

5. Limitations and Future Research

This study is limited by its reliance on bank-specific indicators over a relatively short period (2017–2023) and by the use of only two machine learning algorithms (random forest and SVM). Broader macroeconomic variables, longer time horizons, and alternative models such as gradient boosting or neural networks could provide richer insights. Moreover, the evaluation relied mainly on the RMSE and MAE; including additional metrics and advanced feature selection techniques would enhance robustness. Future research should expand the dataset, apply cross-validation, and test a wider range of methods to improve predictive accuracy and deliver more practical tools for financial institutions and regulators in managing credit risk.

Ethical considerations

Not applicable.

Conflict of interest

The authors declare that they have no conflicts of interest.

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